Decision-OS V6: Canonical Commitment Architecture for Non-Destructive Integration

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Abstract

This paper presents Decision-OS V6 as a minimal architecture for non-destructive commitment. The framework defines how commitment-ready updates are merged, canonicalized, and safety-bounded so that the committed state does not depend on update order, repeated equivalent input, or permissive safety integration. Its core consists of a join-semilattice merge substrate, admissible and inflationary updates, commitment-bounded operational equivalence, idempotent canonicalization, and conservative safety aggregation by max/max/union at the commitment boundary. Under these conditions, V6 guarantees order-robust committed integration, canonical stability under commitment-equivalent repetition, and non-dilution of commitment-relevant safety signals. The framework does not define self-evolution, update selection, temporal optimization, or a full identity theory over the state space. Relative to merge-only formalisms, its contribution is the addition of commitment semantics through bounded canonicalization, bounded operational equivalence, and conservative safety aggregation. While CRDT-style systems explain order-robust convergence of replicated updates, V6 explains the commitment boundary under which such convergence remains non-destructive. V6 should therefore be read as a pre-evolutionary substrate: a minimal structural framework that defines the conditions under which later structural change can occur without destructive inconsistency.

Index Terms

canonical commitment, non-destructive integration, operational equivalence, canonicalization, commitment boundary, conservative safety aggregation, join-semilattice, idempotence

I. Phase-Invariant Memory Architecture (PIC)

A. Scope

Decision-OS V6 is a theory of non-destructive integration at the commitment boundary. It defines how updates are merged, canonicalized, and safety-bounded so that the committed state does not depend on update order. It does not define self-evolution, update selection, or temporal optimization. Its scope is restricted to commitment-safe integration under declared invariants.

B. Core Intuition

V6 treats invariance, not trajectory, as the primary object. Different admissible update paths may produce different intermediate candidates, but they must converge to the same committed canonical state once commitment-relevant distinctions are preserved and commitment-irrelevant distinctions are collapsed. The role of Canon is therefore not global identity reduction, but commitment-preserving collapse. The role of safety is not to suppress candidate exploration, but to prevent destructive inconsistency at commitment.

C. Symbols and Setting

Let S be a state space equipped with a partial order $\leq$ and a join operator $\sqcup$. Let δ denote an update operator on S. Let $\mathrm{Canon} : S \to S$ denote a canonicalization map. Let $\mathcal{Q}_c$ denote the set of commitment-relevant queries defined by the system.

The framework assumes that commitment is evaluated only on distinctions that remain visible through $\mathcal{Q}_c$ and through the commitment decision induced by the same admissible evidence.

D. Axioms

a) A1. Join-semilattice.: $(S, \leq, \sqcup)$ is a join-semilattice.[1], [2] For all $x, y \in S$, the join $x \sqcup y$ exists and is commutative, associative, and idempotent.

b) A2. Admissible update.: An update δ is admissible only if it preserves the declared invariants under join and canonicalization and does not violate commitment-level safety conditions.

For the purposes of V6, declared invariants are system-declared constraints fixed in advance together with the commitment-relevant query set Q_c; V6 does not specify their content.

c) A2'. Inflationary update.: For every admissible δ and every $x \in S$,

$$x \leq \delta(x).$$

d) A3. Canonicalization.: Canon is idempotent:

$$\mathrm{Canon}(\mathrm{Canon}(x)) = \mathrm{Canon}(x).$$

It is not a full identity definition. It collapses only distinctions that are irrelevant to commitment under the declared operational equivalence.

e) A4. Safety Triplet.: Safety is represented at commitment by the triplet

$$(\mathrm{Severity}, \mathrm{Until}, \mathrm{Evidence}),$$

with aggregation rules

$$\mathrm{Severity}^* = \max_t \mathrm{Severity}_t, \qquad \mathrm{Until}^* = \max_t \mathrm{Until}_t, \qquad \mathrm{Evidence}^* = \bigcup_t \mathrm{Evidence}_t\,.$$

The severity order is

$$\mathrm{PASS} < \mathrm{DELAY} < \mathrm{BLOCK}.$$

These aggregation rules apply at commitment, not at candidate generation.

E. Operational Equivalence

Two states $x, y \in S$ are operationally equivalent, written $x \sim_c y$, iff they induce the same outputs for all commitment-relevant queries in $\mathcal{Q}_c$ and the same commitment outcome under the same admissible evidence. The query set is restricted to commitment-relevant queries defined by the system. This equivalence is sufficient for commitment and is not intended as a complete semantic identity relation.

F. Canon Definition

The canonicalization map Canon preserves all distinctions that can change commitment and collapses only distinctions that cannot. Formally, if $x \sim_c y$, then

$$\mathrm{Canon}(x) = \mathrm{Canon}(y).$$

Conversely, if a distinction changes at least one answer in $\mathcal{Q}_c$ or changes the commitment outcome under the same admissible evidence, that distinction must remain visible prior to commitment and must not be collapsed by Canon.

Canon is therefore commitment-preserving rather than representation-erasing.

G. Merge Rule

For admissible updates, committed integration is defined by

$$F_\delta(x) := \mathrm{Canon}(\delta(x)),$$

and pairwise committed merge is defined by

$$M_{\delta_i,\delta_j}(x) := \mathrm{Canon}\big(\delta_i(x) \sqcup \delta_j(x)\big).$$

Because $\sqcup$ is commutative, associative, and idempotent, and because Canon is idempotent, committed integration is order-robust whenever the validity conditions below hold.

H. Propositions

a) P1. Order-independence.: Let δ_i, δ_j be admissible updates. Then the committed state obtained from δ_i and δ_j is invariant under permutation:

$$\mathrm{Canon}\big(\delta_i(x) \sqcup \delta_j(x)\big) = \mathrm{Canon}\big(\delta_j(x) \sqcup \delta_i(x)\big).$$

b) Proof sketch.: By A1, $\sqcup$ is commutative. Applying the same idempotent canonicalization on both sides yields the same committed canonical state.

c) P2. Idempotent stability.: For every admissible update δ,

$$\mathrm{Canon}\big(\mathrm{Canon}(\delta(x))\big) = \mathrm{Canon}(\delta(x)).$$

d) Proof sketch.: This follows directly from the idempotence of Canon in A3.

e) P3. Safety non-dilution.: Let a sequence of commitment-relevant safety signals be aggregated by A4. Then adding further signals cannot weaken the committed safety stance.

f) Proof sketch.: $\mathrm{Severity}^*$ and Until^* are aggregated by max, and $\mathrm{Evidence}^*$ by set union. Each operation is monotone with respect to additional inputs. Hence the committed safety state cannot be diluted by later aggregation.

I. Validity Conditions

The preceding propositions hold only if all of the following conditions are satisfied: A1–A4 and A2′ hold; all updates are admissible; the operational equivalence relation is specified only for commitment-relevant queries and commitment outcomes; and safety aggregation is applied at commitment rather than during candidate generation.

V6 does not claim order-independence, idempotent stability, or safety preservation outside these conditions.

J. Failure Modes

If an inadmissible update is merged, declared invariants may be violated and committed confluence need not hold.

If Canon is over-compressive, commitment-relevant distinctions may be erased, and operational equivalence becomes unsound.

If the Safety Triplet is applied before commitment to candidate states, exploration is prematurely collapsed and the architecture ceases to distinguish frontier generation from committed integration.

If operational equivalence is defined too broadly or against unrestricted queries, Canon becomes a hidden ontology claim rather than a commitment-bounded canonicalization rule.

K. CRDT Comparison

V6 shares with CRDT-style systems an algebraic substrate based on join-semilattice structure and order-robust merge. However, V6 is not reducible to CRDT formulation alone. Its distinctive layer is the combination of commitment-bounded operational equivalence, canonicalization by commitment relevance, and non-dilutive safety aggregation at the commitment boundary. The contribution is therefore not merge alone, but merge plus Canon plus Safety under commitment semantics.

L. Boundary to Evolution

A divergence signal such as $\Delta H > 0$ may be treated as a necessary indication that later layers have non-trivial update potential. However, V6 does not optimize, threshold, or otherwise theorize ΔH. V6 defines only the non-destructive substrate on which such later theories may be built. In this sense, V6 is not a theory of self-evolution. It defines the structural conditions under which self-evolution can occur without destructive inconsistency.

II. Commitment Semantics and Validity Boundary

A. Purpose of the Second Layer

The first layer defined the algebraic core of V6: admissible update, join-based merge, canonicalization, and commitment-level safety aggregation. The second layer fixes how these objects are interpreted at commitment. Its purpose is not to introduce a new theory, but to prevent scope drift in the reading of the first layer.

In particular, this section clarifies four points: what commitment means, what counts as a commitment-preserving distinction, why candidate generation must remain outside conservative merge, and under which validity conditions the V6 guarantees hold.

B. Commitment Boundary

V6 distinguishes between candidate formation and committed state integration.

A candidate state is a provisional structure that may still contain mutually competing descriptions, unresolved alternatives, or distinctions whose commitment relevance has not yet been determined. A committed state is the result of applying admissible merge, canonicalization, and the Safety Triplet under declared invariants.

The architecture therefore enforces the following boundary:

> Conservative merge rules belong to commitment, not to candidate generation.

This distinction is necessary because candidate exploration must temporarily preserve alternatives, while commitment must remove order dependence and safety dilution.

C. Candidate Frontier and Committed Canonical State

Let $\mathcal{F}(x)$ denote a candidate frontier generated from state x. The frontier may contain multiple candidates

$$c_1, c_2, \ldots, c_n,$$

without requiring immediate collapse into a single representative.

Commitment is defined only after admissibility, commitment relevance, and safety conditions are evaluated. At that point, the committed canonical state is produced by

$$x^+ := \mathrm{Canon}\Big(\bigsqcup_{c \in \mathcal{A}(x)} c\Big),$$

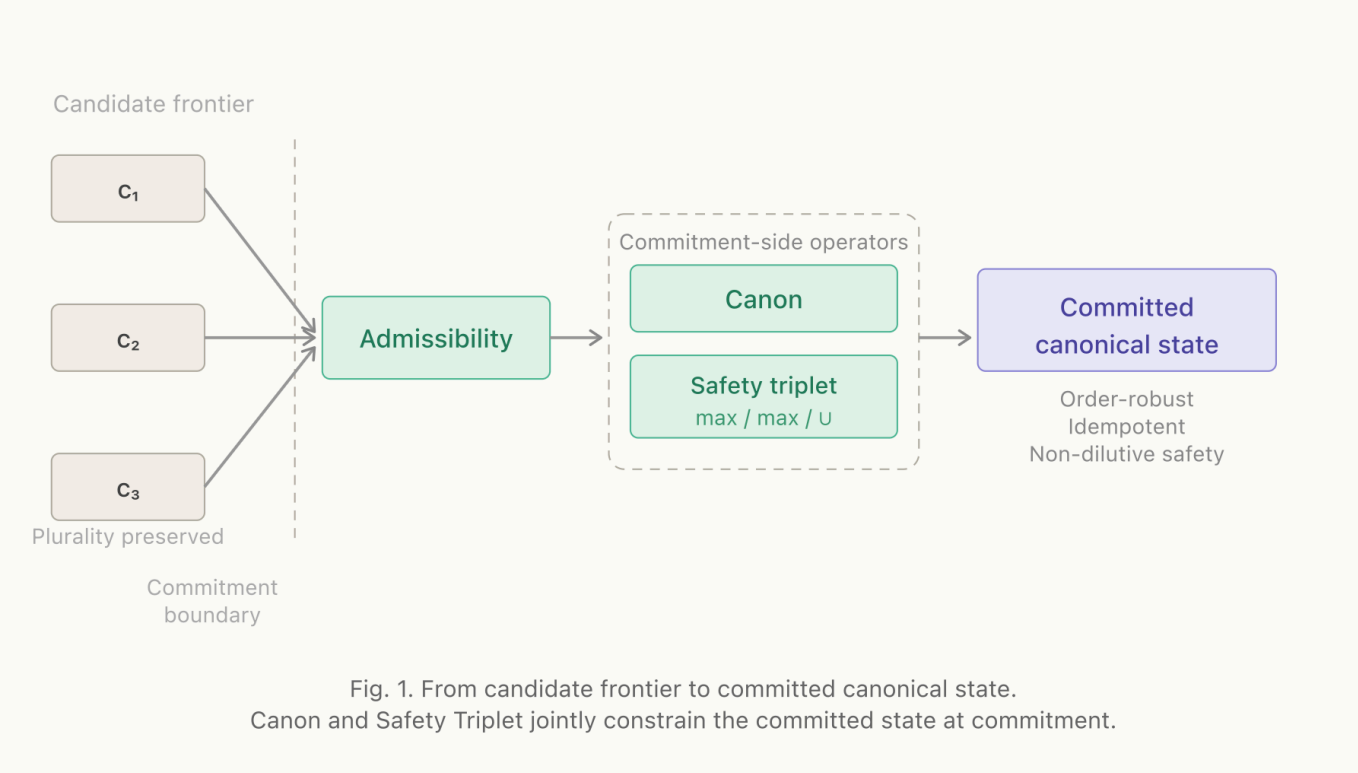

Fig. 1. From candidate frontier to committed canonical state. Canon and Safety Triplet jointly constrain the committed state at commitment.

where $\mathcal{A}(x)$ is the set of admissible commitment-ready candidates.

The frontier is therefore exploratory, whereas the committed state is canonical. V6 governs the second object. It does not require the frontier itself to be canonical.

Figure 1 illustrates the separation between candidate plurality before commitment and the committed canonical state after commitment-side constraints are applied.

D. Why Safety Is Deferred to Commitment

If conservative safety aggregation is applied directly to the candidate frontier, then exploratory alternatives can be eliminated before their commitment relevance is determined. This collapses the distinction between search and decision.

V6 avoids this failure by assigning different roles to the two phases.

At the frontier: alternatives may coexist.

At commitment: the Safety Triplet becomes binding, and the committed state must respect

$$\text{Severity}^* = \max_t \text{Severity}_t, \qquad \text{Until}^* = \max_t \text{Until}_t, \qquad \text{Evidence}^* = \bigcup_t \text{Evidence}_t\,.$$

This design preserves exploration while preventing committed safety signals from being weakened by order or omission.

E. Commitment-Preserving Distinctions

A distinction is commitment-preserving iff removing it changes either: (i) at least one answer to a commitment-relevant query in $\mathcal{Q}_c$, or (ii) the commitment outcome under the same admissible evidence.

A distinction is commitment-irrelevant iff removing it changes neither.

Accordingly, canonicalization is constrained as follows.

a) C1. Preservation condition.: If a distinction is commitment-preserving, Canon must not collapse it before commitment.

b) C2. Collapse condition.: If two states differ only by commitment-irrelevant distinctions, Canon may map them to the same canonical representative.

These two conditions jointly prevent two opposite failures: under-collapse, in which order-dependent redundancy remains in the committed state; and over-collapse, in which commitment-relevant structure is erased.

F. Minimal Commitment Operator

Let Commit denote the commitment operator. For a frontier $\mathcal{F}(x)$, define

$$\mathrm{Commit}(\mathcal{F}(x)) := \mathrm{Canon}\Big(\bigsqcup_{c \in \mathcal{A}(x)} c\Big),$$

subject to: all $c \in \mathcal{A}(x)$ are admissible, the declared invariants are preserved, and the aggregated Safety Triplet does not violate commitment-level constraints.

This operator is minimal in the following sense: it says nothing about how candidates are generated or selected beyond admissibility for commitment. It defines only how commitment-ready structure is integrated without destructive inconsistency.

G. Operational Consequence

The main consequence of the commitment operator is that V6 guarantees are stated only for the committed state, not for the exploratory frontier.

Thus, when V6 claims order-independence, idempotent stability, and safety non-dilution, the claim is always of the form:

$$\text{guarantee} \implies \text{committed canonical state},$$

not:

$$\text{guarantee} \implies \text{all intermediate candidates}.$$

This restriction is essential. Without it, the theory would implicitly govern hypothesis generation, update selection, or temporal control, which are outside its scope.

H. Validity Conditions Revisited

The V6 guarantees are valid only if the following boundary conditions hold simultaneously.

First, the merge substrate must satisfy A1. Second, updates entering commitment must satisfy admissibility and inflationarity. Third, Canon must be interpreted through commitment-bounded operational equivalence rather than global identity. Fourth, safety aggregation must be binding at commitment but not prematurely imposed on the candidate frontier. Fifth, the query set used in operational equivalence must remain restricted to commitment-relevant queries.

If any one of these conditions fails, V6 does not guarantee confluence of commitment.

I. Failure Modes Revisited

A common failure is premature commitment. This occurs when a frontier candidate is treated as if it were already canonical. The result is hidden order dependence, because later admissible candidates no longer enter the merge symmetrically.

A second failure is frontier collapse by safety. This occurs when BLOCK-level or DELAY-level aggregation is applied before commitment relevance is established. The result is suppression of exploratory structure rather than protection of committed structure.

A third failure is unrestricted equivalence. If operational equivalence is defined over arbitrary queries rather than commitment-relevant ones, canonicalization becomes under-justified and can erase distinctions that still matter operationally.

A fourth failure is silent inadmissibility. If a candidate enters commitment without preserving declared invariants, the resulting state may still look canonical syntactically while failing semantically.

J. Relation to Self-Evolution

The commitment boundary explains why V6 is not a theory of self-evolution.

Self-evolution requires at least three additional elements that V6 does not provide: a rule for update generation beyond admissible integration, a criterion for selection among competing future trajectories, and a temporal policy for when structural change should occur.

V6 provides none of these. Its role is narrower and more fundamental: it defines the non-destructive substrate on which such later layers may be built.

K. Chapter Conclusion

The second layer closes the interpretation of the first. V6 does not govern all change. It governs only the integration of commitment-ready updates under admissibility, canonicalization, and non-dilutive safety aggregation.

The result is a precise responsibility boundary: candidate plurality is preserved before commitment, while committed structure is made order-robust, idempotent, and safety-bounded at commitment.

III. Operational Equivalence and Canonical Stability

A. Role of This Layer

The first layer defined the algebraic core of merge and canonicalization. The second layer fixed the commitment boundary and separated candidate plurality from committed integration. This third layer closes the remaining semantic gap: it specifies how canonical stability is grounded by operational equivalence rather than by unrestricted identity.

The purpose of this section is therefore narrow. It does not expand the architecture. It only states when canonical collapse is justified, when it is not, and why the resulting committed state is stable under repeated integration.

B. Why Full Identity Is Unnecessary

V6 does not require a full semantic identity theory over the state space S. Such a requirement would exceed the scope of commitment-safe integration and would force the architecture to answer questions irrelevant to its stated purpose.

Instead, V6 adopts a weaker and operational notion: states need only be equivalent with respect to commitment-relevant queries and commitment outcomes under the same admissible evidence.[3]

This restriction is deliberate. The goal is not to define what two states are in every possible sense. The goal is to define when they can be treated as the same for commitment without destructive inconsistency.

C. Commitment-Bounded Equivalence

Recall the commitment-bounded equivalence relation

$$x \sim_c y.$$

By definition,

$$x \sim_c y$$

iff x and y induce the same answers for all queries in $\mathcal{Q}_c$ and the same commitment outcome under the same admissible evidence.

This relation is bounded in three ways.

First, it is query-bounded: only queries in $\mathcal{Q}_c$ matter.

Second, it is evidence-bounded: comparison is made under the same admissible evidence.

Third, it is decision-bounded: equivalence is judged only relative to commitment outcome.

These restrictions prevent the equivalence relation from silently expanding into a total semantic collapse.

D. Canonicalization as Quotient by Commitment Relevance

Under the bounded relation $\sim_c$, the role of Canon is to map each state to a representative of its commitment-equivalence class.

Formally, for every $x \in S$,

$$\mathrm{Canon}(x) \in [x]_{\sim_c},$$

where $[x]_{\sim_c}$ denotes the equivalence class of x under $\sim_c$.

The purpose of canonicalization is therefore not compression for its own sake. Its purpose is to remove distinctions that are provably irrelevant to commitment while preserving all distinctions that remain commitment-preserving.

Hence the architecture uses quotienting only in a bounded operational sense.

E. Canonical Stability

A committed state is canonically stable iff reapplying admissible integration and canonicalization does not change its commitment-equivalence class.

Let x^* be a committed canonical state. Then canonical stability requires

$$\mathrm{Canon}(x^*) = x^*$$

and, for every admissible update δ that contributes no new commitment-relevant distinction beyond what is already preserved in x^*,

$$\mathrm{Canon}(\delta(x^*)) = x^*.$$

This is the V6 notion of stability. It is not immobility of the entire state space. It is invariance of the committed canonical state under commitment-irrelevant repetition.

F. Idempotence and Repetition

The importance of idempotence in V6 is not merely algebraic convenience. It is what prevents repeated integration from generating artificial difference.

Suppose the same admissible update is merged multiple times. Because join is idempotent and Canon is idempotent,

$$\mathrm{Canon}(x \sqcup x) = \mathrm{Canon}(x),$$

and repeated canonicalization yields no further change:

$$\mathrm{Canon}(\mathrm{Canon}(x)) = \mathrm{Canon}(x).$$

Thus, once a committed structure has been canonically fixed, repetition alone cannot create drift. This property is necessary for any claim that commitment is robust to duplicated updates, reordered arrival, or repeated reconciliation.

G. Proposition: Equivalence-Preserving Canonicalization

a) P4.: If $x \sim_c y$, then

$$\mathrm{Canon}(x) = \mathrm{Canon}(y).$$

b) Proof sketch.: By definition, Canon collapses only distinctions that are irrelevant under $\sim_c$. Since x and y are indistinguishable with respect to all commitment-relevant queries and commitment outcome under the same admissible evidence, they belong to the same commitment-equivalence class. Therefore Canon maps them to the same canonical representative.

H. Proposition: Canonical Stability Under Repetition

a) P5.: Let x^* be a committed canonical state. If an admissible update δ introduces no new commitment-relevant distinction, then

$$\mathrm{Canon}(\delta(x^*)) = x^*.$$

b) Proof sketch.: By assumption, $\delta(x^*)$ remains in the same commitment-equivalence class as x^*. Applying P4 yields identical canonical representatives. Hence the committed state is stable under repetition of commitment-irrelevant admissible integration.

I. What Canonical Stability Does Not Claim

Canonical stability does not claim that every admissible update is absorbed without change.

If an admissible update introduces a new commitment-preserving distinction, then the committed canonical state may change. This is not a failure of stability. It is the intended behavior of commitment-safe integration.

Nor does canonical stability claim that all future updates are already encoded in the present state. It claims only that repeated integration of commitment-equivalent structure does not create artificial divergence.

This distinction is essential. Without it, idempotence would be misread as a prohibition on meaningful update.

J. Bad Equivalence as a Failure Mode

The most direct way to break canonical stability is to define equivalence badly.

If $\sim_c$ is too narrow, trivial representational differences remain uncollapsed and the committed state retains unnecessary order-sensitive redundancy.

If $\sim_c$ is too broad, commitment-relevant distinctions are erased and canonicalization becomes unsound.

If $\sim_c$ is allowed to vary implicitly during evaluation, the same pair of states may be judged equivalent in one passage and non-equivalent in another, destroying the reproducibility of commitment.

For this reason, Q_c, admissible evidence conditions, and commitment criteria must be declared before equivalence is used. In particular, Q_c must not be modified during the same commitment cycle.

K. Relation to Conservative Safety

Operational equivalence does not override the Safety Triplet. The two layers have different roles.

Equivalence determines which distinctions may be collapsed without changing commitment meaning.

Safety determines how commitment-relevant risk signals aggregate once commitment is evaluated.

Accordingly, even if two states are equivalent with respect to ordinary commitment-relevant queries, a newly added safety signal may still change the committed state because severity, until, or evidence has changed under the commitment rule.

This is why V6 combines equivalence-bounded canonicalization with non-dilutive safety aggregation rather than reducing one to the other.

L. Consequence for Reviewer Objections

The bounded formulation answers four standard objections.

First, V6 does not require an ontology of full state identity.

Second, Canon is not an arbitrary compression function.

Third, idempotence does not erase meaningful novelty; it removes only commitment-irrelevant repetition.

Fourth, stability is claimed only at the committed canonical layer, not across unrestricted exploratory generation.

These restrictions are not weaknesses. They are the mechanism by which the theory remains minimal and reviewable.

M. Chapter Conclusion

V6 obtains canonical stability by using commitment-bounded operational equivalence rather than unrestricted identity. The committed state is stable because equivalence-preserving repetition is absorbed by join and canonicalization, while new commitment-relevant distinctions remain able to change the state.

The result is a minimal semantic closure: enough to justify canonicalization, not enough to inflate V6 into a theory of semantics, search, or self-evolution.

IV. Conservative Safety Aggregation at Commitment

A. Role of This Layer

The previous layers defined admissible merge, commitment-bounded canonicalization, and the semantic conditions for canonical stability. This layer specifies the safety rule that becomes binding at commitment.

Its role is limited and explicit. It does not define a general risk theory. It does not govern candidate generation. It defines only how commitment-relevant safety signals are aggregated so that the committed state does not become less safe through reordering, repetition, or partial omission.

B. Why Safety Requires Conservative Aggregation

Commitment is the point at which unresolved structure becomes actionable structure. At that point, safety-relevant signals must not be weakened by update order or by averaging across conflicting observations.

A non-conservative rule would permit the following failure: a severe signal arrives, but later weaker signals numerically dilute it, producing a committed state that appears safer only because aggregation was permissive.

V6 excludes this class of failure. Its safety rule is therefore conservative by construction.

C. Safety Triplet

For commitment-relevant safety observations indexed by t, define the Safety Triplet

$$(\mathrm{Severity}_t, \mathrm{Until}_t, \mathrm{Evidence}_t).$$

The committed aggregate is

$$\mathrm{Severity}^* = \max_t \mathrm{Severity}_t, \qquad \mathrm{Until}^* = \max_t \mathrm{Until}_t, \qquad \mathrm{Evidence}^* = \bigcup_t \mathrm{Evidence}_t\,.$$

The severity order is fixed as

$$\mathrm{PASS} < \mathrm{DELAY} < \mathrm{BLOCK}.$$

The semantics of the three coordinates are distinct.

$\mathrm{Severity}^*$ preserves the strongest committed safety stance.

Until^* preserves the latest active commitment-relevant constraint.

$\mathrm{Evidence}^*$ preserves the total evidential support accumulated across commitment-relevant observations.

The aggregate therefore becomes stricter or equal under additional safety input, never weaker.

D. Why the Operators Are $\max / \max / \cup$

The use of max for severity is required because severity is ordinal rather than metric. PASS, DELAY, and BLOCK are not quantities to be averaged. They are ordered commitment stances. Once a stronger stance is justified, averaging would misrepresent the semantics of the order.

The use of max for until is required because a later valid restriction cannot be safely erased by an earlier one. If one commitment-relevant signal extends the protected interval, the aggregate must preserve that extension.

The use of set union for evidence is required because commitment safety depends on retained support, not compressed support. Evidence must accumulate monotonically unless explicit evidence invalidation is defined outside the present theory. V6 does not define such invalidation rules.

E. Safety Monotonicity

The Safety Triplet is monotone with respect to additional commitment-relevant observations.

If a new safety observation is added, then:

$$\text{Severity}^*_{\text{new}} \geq \text{Severity}^*_{\text{old}}, \qquad \text{Until}^*_{\text{new}} \geq \text{Until}^*_{\text{old}}, \qquad \text{Evidence}^*_{\text{old}} \subseteq \text{Evidence}^*_{\text{new}}\,.$$

Thus the committed safety state is monotone in all three coordinates. This property is the basis of safety non-dilution.

F. Proposition: Safety Non-Dilution

a) P6.: Let Σ be a finite set of commitment-relevant safety observations, and let $\Sigma' \supseteq \Sigma$. Then the aggregate Safety Triplet computed from Σ' is never weaker than the one computed from Σ.

b) Proof sketch.: Severity is aggregated by max over a fixed ordinal order, so adding observations cannot decrease the result. Until is aggregated by max over the same order on admissible deadlines, so adding observations cannot shorten the aggregate bound. Evidence is aggregated by set union, so adding observations cannot remove prior evidence. Hence the aggregate under Σ' weakly dominates the aggregate under Σ in all three coordinates.

G. Why Safety Applies at Commitment, Not at the Frontier

The same conservative rule cannot be imposed directly on exploratory candidates without collapsing the distinction between possibility and commitment.

At the candidate frontier, incompatible hypotheses may need to coexist. A max-style safety collapse at that stage would turn one strong local warning into a global frontier-wide veto before commitment relevance is evaluated.

This is too strong for exploration and too weak for commitment semantics, because it confuses search control with committed safety state construction.

V6 therefore imposes the following separation:

frontier generation may preserve plurality;

commitment aggregation must preserve conservative safety.

The Safety Triplet is binding only in the second phase.

H. Commitment Interaction with Canon

Safety aggregation and canonicalization do not duplicate each other.

Canonicalization collapses distinctions that are irrelevant to commitment under bounded operational equivalence.

Safety aggregation preserves commitment-relevant risk structure once commitment is evaluated.

The interaction rule is therefore: canonicalization may simplify only after commitment-relevant safety distinctions have been preserved in the committed state.

Equivalently, Canon cannot erase a distinction whose removal would change Severity*, Until*, or Evidence* at commitment.

I. Proposition: Canon Cannot Dilute Committed Safety

a) P7.: Let x and y be states such that replacing x by y changes at least one coordinate of the committed Safety Triplet. Then $x \not\sim_c y$, and Canon must not collapse them to the same committed representative.

b) Proof sketch.: Operational equivalence is defined partly by preservation of commitment outcome under the same admissible evidence. A change in committed safety aggregation changes commitment semantics. Therefore the two states are not commitment-equivalent. By the preservation rule for Canon, they cannot be collapsed into the same committed representative.

J. Failure of Averaging

Averaging is not a harmless alternative to conservative aggregation.

If severity is averaged, a BLOCK signal can be numerically diluted by multiple PASS signals. If until is averaged, a longer valid restriction can be shortened by unrelated shorter ones. If evidence is compressed rather than accumulated, commitment may lose the trace needed to justify the committed stance.

For these reasons, V6 excludes averaging from the commitment layer. This is not an implementation preference. It is a structural requirement for non-destructive commitment.

K. Residual Limits

The Safety Triplet is intentionally minimal.

It does not define how safety observations are generated. It does not define evidential trust scoring. It does not define invalidation, decay, or source reliability weighting. It does not define what policy should follow once BLOCK is reached.

Those are downstream design questions. V6 defines only the conservative aggregation rule required so that commitment does not become less safe through integration.

L. Reviewer-Facing Boundary

This bounded formulation answers three predictable objections.

First, V6 does not claim to solve general safety. It defines only conservative safety aggregation at commitment.

Second, the triplet is not an ad hoc heuristic. Its operators are forced by the requirement that commitment safety not be weakened by reordering or partial accumulation.

Third, the theory does not kill exploration, because the triplet is binding only at commitment and not at candidate generation.

These limits are part of the contribution. They keep the safety layer minimal, precise, and auditable.

M. Chapter Conclusion

The Safety Triplet supplies the commitment-side conservative rule of V6. Its aggregation by

$$\max / \max / \cup$$

ensures that committed safety cannot be diluted by update order, repetition, or permissive merge.

This does not make V6 a full theory of risk. It makes V6 a theory of non-destructive commitment under conservative safety aggregation.

V. Structural Propositions and Minimal Proof Sketches

A. Purpose of This Layer

The previous layers defined the merge substrate, the commitment boundary, commitment-bounded operational equivalence, canonical stability, and conservative safety aggregation. This section collects the main structural consequences into a compact proposition layer.

The purpose is not to enlarge the theory. It is to make explicit which claims V6 actually proves, which assumptions they depend on, and how far those claims extend.

B. Standing Assumptions

Throughout this section, we assume A1–A4 and A2′. In particular, A2′ excludes update rules that reduce previously preserved structure during commitment integration. We further assume that: all updates entering commitment are admissible, the query set Q_c is fixed prior to commitment evaluation, is commitment-relevant, and must not be modified during the same commitment cycle, operational equivalence is evaluated under the same admissible evidence, and the Safety Triplet is applied at commitment rather than at candidate generation.

All propositions below are conditional on these assumptions. If any one of them fails, the propositions need not hold.

C. Proposition 1: Order-Independence of Committed Merge

a) P1.: Let δ_i and δ_j be admissible updates. Then the committed merge is invariant under permutation:

$$\mathrm{Canon}\big(\delta_i(x) \sqcup \delta_j(x)\big) = \mathrm{Canon}\big(\delta_j(x) \sqcup \delta_i(x)\big).$$

b) Proof sketch.: By A1, the join $\sqcup$ is commutative. Applying the same idempotent canonicalization to both sides yields the same committed representative. Hence permutation does not change the committed canonical state.

D. Proposition 2: Associative Robustness of Committed Merge

a) P2.: Let $\delta_1, \delta_2, \delta_3$ be admissible updates. Then

$$\mathrm{Canon}\big((\delta_1(x) \sqcup \delta_2(x)) \sqcup \delta_3(x)\big) = \mathrm{Canon}\big(\delta_1(x) \sqcup (\delta_2(x) \sqcup \delta_3(x))\big).$$

b) Proof sketch.: By A1, $\sqcup$ is associative. Canonicalization is applied only after the same total join is formed. Therefore the grouping of admissible committed updates does not affect the resulting committed canonical state.

E. Proposition 3: Idempotent Stability of Canonicalization

a) P3.: For every state $x \in S$,

$$\mathrm{Canon}(\mathrm{Canon}(x)) = \mathrm{Canon}(x).$$

b) Proof sketch.: This is A3. Its importance is operational: once a committed representative has been obtained, repeated canonicalization alone cannot generate additional change.

F. Proposition 4: Equivalence-Preserving Canonicalization

a) P4.: If $x \sim_c y$, then

$$\mathrm{Canon}(x) = \mathrm{Canon}(y).$$

b) Proof sketch.: By construction, Canon collapses only distinctions irrelevant under commitment-bounded operational equivalence. Thus states in the same commitment-equivalence class map to the same committed representative.

G. Proposition 5: Canonical Stability Under Commitment-Irrelevant Repetition

a) P5.: Let x^* be a committed canonical state. If an admissible update δ introduces no new commitment-preserving distinction, then

$$\mathrm{Canon}(\delta(x^*)) = x^*.$$

b) Proof sketch.: Under the stated condition, $\delta(x^*)$ remains in the same commitment-equivalence class as x^*. Applying P4 yields the same committed representative. Hence repetition of commitment-irrelevant admissible structure does not induce drift.

H. Proposition 6: Safety Non-Dilution

a) P6.: Let Σ be a finite set of commitment-relevant safety observations, and let $\Sigma' \supseteq \Sigma$. Then the aggregate Safety Triplet computed from Σ' is never weaker than the one computed from Σ.

b) Proof sketch.: Severity and until are aggregated by max, and evidence by set union. Each operator is monotone with respect to additional inputs. Therefore adding commitment-relevant safety observations cannot weaken the committed safety state.

I. Proposition 7: Canon Cannot Dilute Committed Safety

a) P7.: Let x and y be states such that replacing x by y changes at least one coordinate of the committed Safety Triplet. Then $x \not\sim_c y$, and Canon must not collapse them to the same committed representative.

b) Proof sketch.: A change in committed severity, until, or evidence changes commitment meaning under the same admissible evidence. Therefore the two states are not commitment-equivalent. By the preservation condition on canonicalization, they cannot be mapped to the same committed representative.

J. Proposition 8: Commitment Confluence

a) P8.: Let $\Delta = \{\delta_1, \ldots, \delta_n\}$ be a finite set of admissible commitment-ready updates. Then the committed canonical state

$$x^+ := \mathrm{Canon}\Big(\bigsqcup_{k=1}^{n} \delta_k(x)\Big)$$

is independent of the order of enumeration of Δ.

b) Proof sketch.: By induction on $|\Delta|$: the base case $n = 2$ is P1. For the inductive step, assume the result for $n = k$; then for $n = k + 1$, associativity and commutativity of $\sqcup$ allow the additional admissible update to be merged without changing the joined state up to permutation, and idempotent canonicalization yields the same committed representative. Thus the committed state depends only on the admissible set entering commitment, not on the order in which it arrived.

K. What These Propositions Establish

Taken together, P1–P8 establish the following limited result.

V6 guarantees confluence of committed integration under admissibility, bounded operational equivalence, and conservative safety aggregation. This confluence has three components:

order-independence, idempotent stability, and safety non-dilution.

These are the entire core guarantees of V6. No stronger result is claimed in this paper.

L. What These Propositions Do Not Establish

These propositions do not establish a theory of search. They do not establish a theory of self-evolution. They do not establish a criterion for generating candidates, choosing among future trajectories, or optimizing temporal divergence.

They also do not establish a complete semantics of the state space S. The equivalence relation used here is deliberately bounded by commitment relevance.

Accordingly, the present proposition layer is structural rather than global. It proves exactly what is needed for non-destructive commitment and nothing beyond that.

M. Dependence on Validity Conditions

The proposition layer is only as strong as its validity conditions.

If inadmissible updates enter commitment, confluence may fail.

If the query set $\mathcal{Q}_c$ is not fixed, equivalence may shift during evaluation and canonicalization may become unstable.

If safety aggregation is moved from commitment to frontier generation, exploration and commitment are conflated and the safety propositions no longer describe the intended architecture.

If Canon is allowed to erase commitment-preserving distinctions, then the proofs become unsound because operational equivalence is no longer preserved.

Thus, the assumptions are not formatting details. They are constitutive parts of the theorem boundary.

N. Reviewer-Facing Interpretation

A likely reviewer concern is that the proposition layer appears modest. That modesty is intentional.

The contribution of V6 is not to solve all update semantics. It is to isolate a minimal structural core under which commitment can be made order-robust, canonically stable, and safety non-dilutive.

Any broader claim would require additional theory not given here. By refusing those broader claims, V6 remains sharper rather than weaker.

O. Chapter Conclusion

The proposition layer closes the formal core of V6. Under the stated assumptions, admissible commitment-ready updates converge to a committed canonical state that is invariant under order, stable under repetition of commitment-equivalent structure, and protected against safety dilution.

This is the maximal result claimed by V6. It is also the precise boundary that keeps the theory minimal and reviewable.

VI. Failure Modes and Boundary Cases

A. Purpose of This Layer

The previous sections defined the scope, axioms, commitment boundary, operational equivalence, conservative safety aggregation, and the resulting structural propositions. This section states how the framework fails when its constitutive conditions are violated.

Its purpose is not defensive rhetoric. It is part of the theory. A minimal framework is reviewable only if its failure boundary is explicit.

B. Failure Is a Boundary Condition, Not an Exception

V6 makes conditional guarantees. It does not claim that non-destructive commitment holds under arbitrary update rules, arbitrary equivalence relations, or arbitrary safety procedures.

Accordingly, failure in V6 does not mean that the framework is incoherent. It means that one or more validity conditions have been violated.

The task of this section is therefore to identify the smallest set of violations that break confluence, canonical stability, or safety non-dilution.

C. Failure Mode 1: Inadmissible Update

The first failure mode occurs when an update enters commitment without satisfying admissibility.

Formally, if an update δ does not preserve declared invariants under join and canonicalization, or if it violates commitment-level safety conditions, then δ is inadmissible.

Once such an update is merged, V6 no longer guarantees that the committed state is well-formed. The resulting state may still appear syntactically integrated, but confluence is no longer justified.

This failure is fundamental because admissibility is not an optional filter. It is a constitutive entry condition for commitment.

D. Failure Mode 2: Over-Compression by Canon

The second failure mode occurs when Canon collapses a distinction that is still commitment-preserving.

If two states differ in a way that changes at least one answer in $\mathcal{Q}_c$ or changes the commitment outcome under the same admissible evidence, then they must not be identified by Canon.

If they are identified anyway, the framework loses soundness. The committed representative no longer preserves commitment meaning, and the canonical state becomes an artifact of excessive collapse rather than a valid quotient by commitment relevance.

This is the most direct failure of canonicalization.

E. Failure Mode 3: Under-Compression

The opposite failure also exists.

If Canon is too weak, states that are operationally equivalent under commitment are left distinct. The result is that the committed layer retains unnecessary representational residue.

This does not immediately destroy soundness, but it weakens confluence in practice by making the committed state sensitive to irrelevant encoding differences. Order-independent merge remains harder to observe because redundant distinctions are preserved without need.

Thus V6 excludes both over-compression and under-compression. Canonicalization must be neither stronger nor weaker than commitment relevance.

F. Failure Mode 4: Safety Applied Before Commitment

The fourth failure mode occurs when the Safety Triplet is applied directly to the candidate frontier rather than to the committed state.

At the frontier, alternatives may still be exploratory. A strong local safety signal may be informative without yet determining commitment.

If conservative aggregation is imposed too early, exploratory plurality is collapsed into premature veto. The architecture then ceases to distinguish candidate generation from commitment.

This is not a minor implementation mistake. It changes the theory itself by replacing commitment-safe integration with frontier-wide suppression.

G. Failure Mode 5: Bad Operational Equivalence

The fifth failure mode occurs when operational equivalence is specified badly.

If the query set is too broad, Canon becomes under-justified and may erase distinctions that remain operationally meaningful.

If the query set is too narrow, trivial representational differences remain uncollapsed and the committed state retains irrelevant redundancy.

If the query set changes implicitly during evaluation, reproducibility fails because the same pair of states may be treated as equivalent in one passage and non-equivalent in another.

For this reason, $\mathcal{Q}_c$ is not a convenience. It is part of the formal boundary of the theory.

H. Failure Mode 6: Evidence Mismatch

Operational equivalence is defined under the same admissible evidence. If equivalence is judged under one evidence basis and commitment under another, the canonicalization rule becomes unstable.

Two states may appear equivalent under incomplete evidence yet diverge once full admissible evidence is considered. If collapse occurs before this mismatch is resolved, the committed state becomes unsound.

Hence evidence-bounded comparison is not secondary. It is required to keep equivalence and commitment aligned.

I. Failure Mode 7: Order-Sensitive Merge Rule

If the merge substrate is not a join-semilattice, or if the effective merge procedure violates commutativity, associativity, or idempotence, then commitment confluence fails directly.

In such a case, the same admissible update set may yield different committed states under different arrival orders or grouping patterns.

This is precisely the class of failure V6 is designed to exclude. Without order-robust merge, there is no phase-invariant commitment.

J. Failure Mode 8: Non-Inflationary Update

If an update entering commitment is not inflationary, then previously preserved structure may be reduced rather than extended under integration. In that case, the update sequence need not form a monotone chain, and the convergence intuition underlying committed stability is no longer justified. For this reason, $A2'$ is not a cosmetic strengthening; it is part of the structural boundary of non-destructive commitment.

K. Failure Mode 9: Hidden Policy Injection

Another boundary failure occurs when downstream policy is silently injected into the core layer.

Examples include: using Canon to implement undisclosed selection criteria, treating severity aggregation as if it contained a full risk policy, or interpreting commitment stability as if it implied temporal optimization.

These additions may be useful elsewhere, but they are not part of V6. If inserted without declaration, they inflate the theory and blur the boundary between structural substrate and downstream governance.

This failure is conceptual rather than algebraic, but it matters equally for reviewability.

L. Boundary Case 1: Meaningful Update

Not every change in the committed state is a failure.

If an admissible update introduces a new commitment-preserving distinction, then the committed canonical state may change. This is expected.

V6 does not aim to keep the state fixed under all updates. It aims to ensure that any such change is caused by commitment-relevant structure rather than by order, duplication, or unsafe collapse.

Thus change itself is not evidence against canonical stability.

M. Boundary Case 2: Exploratory Conflict

Conflict at the candidate frontier is also not a failure.

The frontier is allowed to contain competing candidates, mutually incompatible descriptions, or unresolved alternatives. V6 does not require those structures to be canonical before commitment.

Only once commitment is invoked must the admissible subset be integrated into a committed canonical state. Therefore exploratory non-confluence at the frontier does not contradict commitment confluence.

N. Boundary Case 3: Stronger Safety Signal

A newly added stronger safety signal may change the committed state even when other commitment-relevant queries remain unchanged.

This too is not a failure. Safety aggregation is itself part of commitment meaning.

Accordingly, a change in

$$\text{Severity}^*, \quad \text{Until}^*, \quad \text{or} \quad \text{Evidence}^*$$

is sufficient to block collapse under $\sim_c$ and to justify a different committed representative.

O. Boundary Case 4: Repeated Equivalent Input

If the same admissible and commitment-equivalent input is added repeatedly, the committed state should remain unchanged.

This is not merely a convenience property. It is the operational manifestation of idempotent stability.

Failure here would indicate either bad equivalence, hidden inadmissibility, or a non-idempotent implementation of the merge layer.

P. Minimal Diagnostic Table in Prose

The failure structure of V6 can be summarized informally as follows.

If confluence fails, inspect the merge substrate or admissibility.

If canonical stability fails under equivalent repetition, inspect equivalence or canonicalization.

If committed safety weakens after more safety observations are added, inspect whether aggregation still follows

$$\max / \max / \cup .$$

If exploration is prematurely killed, inspect whether safety has been moved from commitment to frontier generation.

These diagnostics are not external debugging advice. They are direct consequences of the theoretical decomposition.

Q. Why Failure Modes Strengthen the Theory

A reviewer may ask why a minimal paper should spend space on failure. The answer is that failure conditions are part of minimality.

Without them, V6 would read as if its guarantees held universally. By stating exactly how the theory breaks, the paper shows that its claims are conditional, bounded, and falsifiable.

This improves precision rather than weakening ambition.

R. Chapter Conclusion

V6 fails in a small number of identifiable ways: inadmissible updates, bad canonicalization, misplaced safety aggregation, bad operational equivalence, evidence mismatch, order-sensitive merge, and non-inflationary update.

These are not peripheral exceptions. They are the exact negations of the conditions under which non-destructive commitment is possible.

By making them explicit, the framework becomes sharper: minimal in scope, conditional in claim, and reviewable at its true boundary.

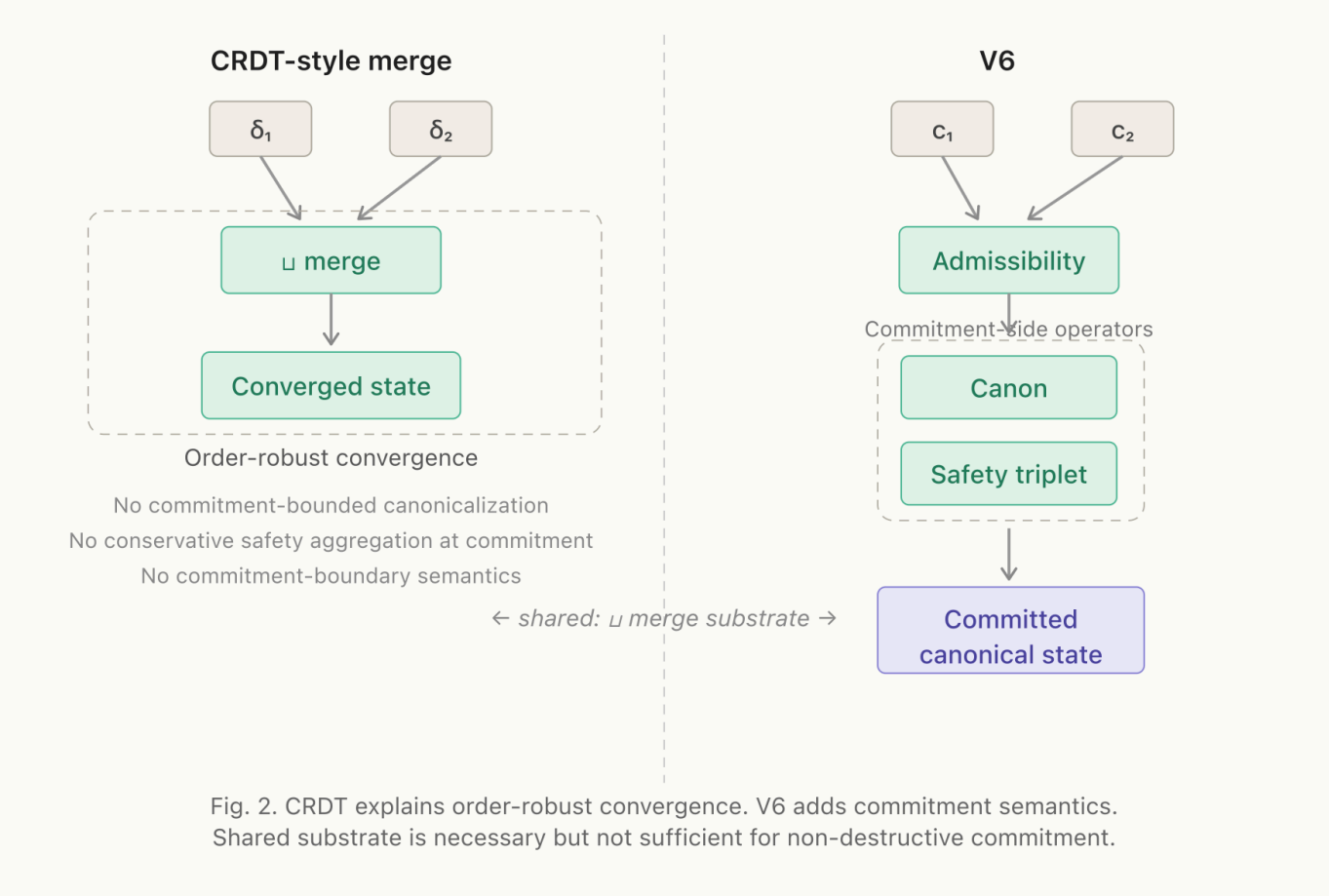

Fig. 2. CRDT explains order-robust convergence. V6 adds commitment semantics. Shared substrate is necessary but not sufficient for non-destructive commitment.

VII. CRDT Comparison and Boundary to Evolution

A. Purpose of This Layer

The preceding sections defined the algebraic core of V6, the commitment boundary, commitment-bounded operational equivalence, conservative safety aggregation, the main structural propositions, and their failure modes. This section places the framework relative to adjacent merge formalisms and closes the scope of the paper at the correct boundary.

Its purpose is narrow. It does not introduce a new layer. It only states what V6 shares with CRDT-style merge systems, what it adds beyond them, and why this still does not make V6 a theory of self-evolution.

B. Shared Substrate with CRDT-Style Systems

V6 shares an algebraic substrate with CRDT-style systems.[1], [2]

Both rely on order-robust integration built on a join-like merge structure. Both require that repeated integration not destabilize the resulting state. Both aim to prevent harmful dependence on message or update arrival order.

At this level, the overlap is real and should be stated plainly. If one asks only whether V6 uses an order-robust, idempotent, merge-compatible substrate, the answer is yes.

Figure 2 summarizes the comparative claim of this paper: CRDT-style systems explain order-robust convergence, whereas V6 explains the commitment boundary under which such convergence remains non-destructive.

C. Why V6 Is Not Reducible to CRDT Alone

The overlap, however, is not the whole theory.

CRDT-style merge typically addresses convergence of replicated state under distributed update arrival. V6 addresses non-destructive commitment under commitment-bounded equivalence and conservative safety aggregation.

The difference is not cosmetic. V6 adds three constraints that are not contained in merge algebra alone.

First, canonicalization is bounded by commitment relevance rather than by representation normalization alone.

Second, equivalence is operational and commitment-scoped rather than unrestricted.

Third, safety aggregation is constitutive at commitment and must remain non-dilutive under integration.

Accordingly, V6 is not reducible to merge convergence alone. Its object is not merely replicated state agreement. Its object is committed canonical state under bounded equivalence and conservative safety.

D. Minimal Comparative Statement

The comparison can therefore be stated compactly as follows.

V6 shares with CRDT-style systems a join-compatible, idempotent, order-robust merge substrate.

V6 differs by adding: commitment-bounded canonicalization, commitment-bounded operational equivalence, and conservative safety aggregation at commitment.

This is the entire comparative claim. No broader historical or implementation claim is needed here.

E. Why Canon Matters Beyond Merge

A pure merge substrate can ensure that multiple updates converge to a common integrated state. That alone does not determine which distinctions may be safely collapsed at commitment.

V6 requires an additional layer because commitment is not only a matter of integration but also of relevance. Two integrated states may still differ only by distinctions that do not matter to commitment. Without Canon, those distinctions remain and the committed state retains unnecessary residue. With over-aggressive Canon, commitment-preserving structure is erased.

Thus merge alone is insufficient. Canon is the bounded rule that closes the gap between raw integration and committed representation.

F. Why Safety Matters Beyond Merge

Likewise, merge alone does not determine how safety-relevant signals should behave at commitment.

A merge substrate may preserve order-robustness while still allowing safety semantics to be weakened by permissive aggregation. V6 excludes this by fixing the Safety Triplet at commitment:

$$\mathrm{Severity}^* = \max_t \mathrm{Severity}_t, \qquad \mathrm{Until}^* = \max_t \mathrm{Until}_t, \qquad \mathrm{Evidence}^* = \bigcup_t \mathrm{Evidence}_t\,.$$

This is not derivable from join algebra alone. It is an additional commitment-side constraint required for non-destructive safety integration.

G. Reviewer-Facing Comparative Boundary

A likely reviewer question is whether V6 is simply a CRDT restatement with new terminology. The answer is no, but the answer should remain narrow.

If one removes Canon, bounded equivalence, and commitment-side safety aggregation, one loses the specific object defined by V6. What remains is only an order-robust merge substrate. That substrate is necessary but not sufficient for the present theory.

Thus the comparison should not be framed as opposition. It should be framed as extension by commitment semantics.

H. Boundary to Evolution

The scope of V6 ends before self-evolution begins.

A later theory of self-evolution would require at least: a rule for generating candidate updates over time, a criterion for selecting among competing future trajectories, and a temporal or dynamical account of when structural change should propagate.

V6 provides none of these. Its guarantees concern only the non-destructive integration of commitment-ready updates.

Accordingly, V6 is not a theory of self-evolution. It defines the structural conditions under which self-evolution can occur without destructive inconsistency.

I. Minimal Bridge Condition

A later layer may require a non-trivial divergence condition in order for structural evolution to be meaningful. In that minimal sense, one may state a bridge condition such as

$$\Delta H > 0.$$

In the present paper, this expression has only boundary status. It says that later layers require some non-zero structural divergence in order to matter.

V6 does not optimize this quantity. It does not threshold it. It does not assign it a preferred operating value. It does not derive any dynamical law from it.

Thus the bridge is acknowledged without being absorbed.

J. Why the Bridge Must Remain Minimal

If the bridge condition were expanded into threshold design, temporal control, or fluctuation optimization, the scope of V6 would immediately inflate beyond commitment-safe integration.

That would recreate the very problem this paper is designed to avoid: a framework that begins as a structural substrate but silently expands into evolution theory.

The bridge must therefore remain one-directional. V6 can support later theories. It does not contain them.

K. Chapter Conclusion

V6 shares with CRDT-style systems a join-compatible, idempotent, order-robust merge substrate, but it is not exhausted by that substrate. Its distinctive contribution is the combination of commitment-bounded canonicalization, commitment-bounded operational equivalence, and conservative safety aggregation at commitment.

This contribution remains strictly pre-evolutionary. V6 does not theorize self-evolution. It defines the non-destructive substrate on which such later theories may be built.

VIII. Conclusion

Decision-OS V6 defines a minimal structural framework for non-destructive commitment. Its purpose is to guarantee that commitment-ready updates can be merged, canonicalized, and safety-bounded without destructive inconsistency arising from order, duplication, or permissive safety integration.

The core claim is narrow and exact. Under admissibility, join-based merge, commitment-bounded operational equivalence, idempotent canonicalization, and conservative safety aggregation at commitment, the committed state is order-robust, canonically stable under commitment-equivalent repetition, and non-dilutive with respect to safety-relevant aggregation.

The framework does not define self-evolution. It does not define update generation, update selection, temporal optimization, or a full identity theory over the state space. Its scope ends at the commitment boundary.

Relative to merge-only formalisms, the contribution of V6 is the addition of commitment semantics: canonicalization bounded by commitment relevance, operational equivalence bounded by commitment-relevant queries and commitment outcome, and conservative safety aggregation by

$$\max / \max / \cup$$

at commitment.

The framework also defines its own failure boundary. If inadmissible updates enter commitment, if Canon collapses commitment-preserving distinctions, if safety is applied prematurely to the candidate frontier, or if operational equivalence is specified badly, the guarantees no longer hold. These are not peripheral exceptions; they are the exact boundary conditions of the theory.

In this sense, V6 should be read neither as a full theory of semantics nor as a theory of self-evolution. It should be read as a pre-evolutionary substrate: a minimal commitment architecture that makes later theories of structural change possible without first permitting destructive inconsistency.

A later layer may require a non-trivial divergence condition such as

$$\Delta H > 0,$$

but that condition lies beyond the present paper. V6 does not optimize, threshold, or otherwise theorize such quantities. It defines only the structural conditions under which later change, if introduced, need not rupture commitment.

The contribution of V6 is therefore subtractive rather than expansive. It removes unnecessary dependence on order, unnecessary dependence on unrestricted identity, and unnecessary weakening of safety at commitment. What remains is a minimal, reviewable theory of canonical commitment.

References

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